# Case 2: Structural Vector AutoRegressive Models, Local Projections and Macro-economic Policy Analysis

EBC4008 – Time Series Analytics

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# Table of Contents

1. Structural VARs
2. Impulse Response Functions for Recursive SVARs
3. VAR/LP IRFs Comparison
4. Structural IRFs Using Local Projections
5. Shock Proxies
6. Final Recommendations

# Recap – Part I

- We estimated a reduced-form VAR(14) for US industrial production, consumer prices and the federal funds rate:

$$y_t = (\Delta \log IP_t,\ \Delta^2 \log CPI_t,\ \Delta FFR_t)' .$$

- We computed reduced-form impulse responses over 48 months, including bootstrap confidence intervals, and accumulated the responses to recover the original variables in levels
- Following a positive FFR innovation, industrial production declined while the CPI level increased, suggesting a **price-puzzle-like response**
- However, a reduced-form FFR innovation is a forecast error, not necessarily an exogenous monetary-policy shock

$\Rightarrow$ **Can structural identification provide a more credible interpretation?**

# Structural VARs

# Reduced-Form VAR versus Structural VAR

Reduced-form VAR: describes dynamic statistical relationships using observable variables and forecast errors

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t$$

Structural VAR: introduces contemporaneous relationships between variables and economically meaningful disturbances

$$B_0 y_t = b + B_1 y_{t-1} + \cdots + B_p y_{t-p} + \varepsilon_t$$

The two representations are linked through:

$$A_i = B_0^{-1} B_i, \qquad u_t = B_0^{-1} \varepsilon_t$$

**Objective:** Identify and isolate economically meaningful structural shocks from reduced-form innovations that may combine multiple disturbances

# From Reduced-Form Errors to Structural Shocks

Let $P = B_0^{-1}$ denote the structural impact matrix. Then:

$$u_t = P\varepsilon_t$$

Assume the structural shocks are orthogonal and have unit variance:

$$E(\varepsilon_t \varepsilon_t') = I_K$$

Therefore, the covariance matrix of reduced-form forecast errors satisfies:

$$\Sigma_u = PP'$$

# From Reduced-Form Errors to Structural Shocks

**Economic interpretation:**

- Each column of $P$ describes how one structural shock affects all variables contemporaneously
- The reduced-form VAR estimates $\Sigma_u$, but does not uniquely determine $P$
- Different impact matrices can imply the same reduced-form covariance matrix

$\Rightarrow$ **Additional identifying restrictions** are required

# Identification Problem

- The impact matrix $P$ contains $K^2$ unknown coefficients
- The symmetric reduced-form covariance matrix $\Sigma_u$ supplies only $\frac{K(K+1)}{2}$ independent covariance equations
- Hence the minimum number of additional independent restrictions for exact identification is:

$$K^2 - \frac{K(K+1)}{2} = \frac{K(K-1)}{2}$$

# Identification Problem

For our three-variable system:

$$\underbrace{9}_{\text{unknowns}} - \underbrace{6}_{\text{covariance equations}} = \boxed{3 \text{ restrictions}}$$

- Covariance information alone does not distinguish economically different shocks
- Restrictions must be independent and satisfy the appropriate rank condition

$\Rightarrow$ The identifying assumptions are therefore part of the economic model, not merely an estimation technique

# Types of Identification Restrictions

Structural shocks can be identified using restrictions motivated by economic theory, institutional information or observed market behavior

**Short-run restrictions:** Impose contemporaneous zero, equality or proportionality restrictions

- Example: Output and prices cannot respond within the month to an unexpected policy decision

**Long-run restrictions:** Restrict the permanent or accumulated impact of structural shocks

- Example: Certain nominal shocks may have no permanent effect on real output

**Sign restrictions:** Restrict the direction of responses over specified horizons rather than setting effects exactly to zero

$\Rightarrow$ **Here: Recursive SVAR using short-run zero restrictions**

# Impulse Response Functions for Recursive SVARs

# Recursive Identification through Ordering

A recursive SVAR imposes a sequential contemporaneous ordering on the variables. For $y_t = (IP_t, CPI_t, FFR_t)'$, its impact matrix is:

$$P = \begin{pmatrix} p_{11} & 0 & 0 \\ p_{21} & p_{22} & 0 \\ p_{31} & p_{32} & p_{33} \end{pmatrix}$$

- The three zero entries supply the restrictions needed for our system
- The Cholesky decomposition selects the lower-triangular $P$ satisfying $\Sigma_u = PP'$
- Structural IRFs are obtained from the reduced-form propagation matrices:

$$\Theta_h = \Phi_h P, \qquad h = 0, \dots, 48$$

$\Rightarrow$ Changing the ordering changes the identifying assumptions and can change the structural responses

# Ordering the Monetary-Policy VAR

We identify monetary-policy shocks with the following recursive ordering:

$$IP_t \longrightarrow CPI_t \longrightarrow FFR_t.$$

**Industrial production (first):** Does not respond on impact to CPI or FFR shocks; real activity adjusts with a delay

**Consumer prices (second):** May respond to IP shocks immediately, but not to monetary-policy shocks

**Federal funds rate (last):** May respond to IP and CPI innovations within the month; its remaining innovation is interpreted as a policy shock

$\Rightarrow$ Limitation: The interpretation depends on credible timing restrictions

# Estimating Recursive SVAR Impulse Responses

Estimate SVAR(14) identifying structural shocks through the Cholesky decomposition:

$$\Theta_h = \Phi_h P, \qquad \Sigma_u = PP'$$

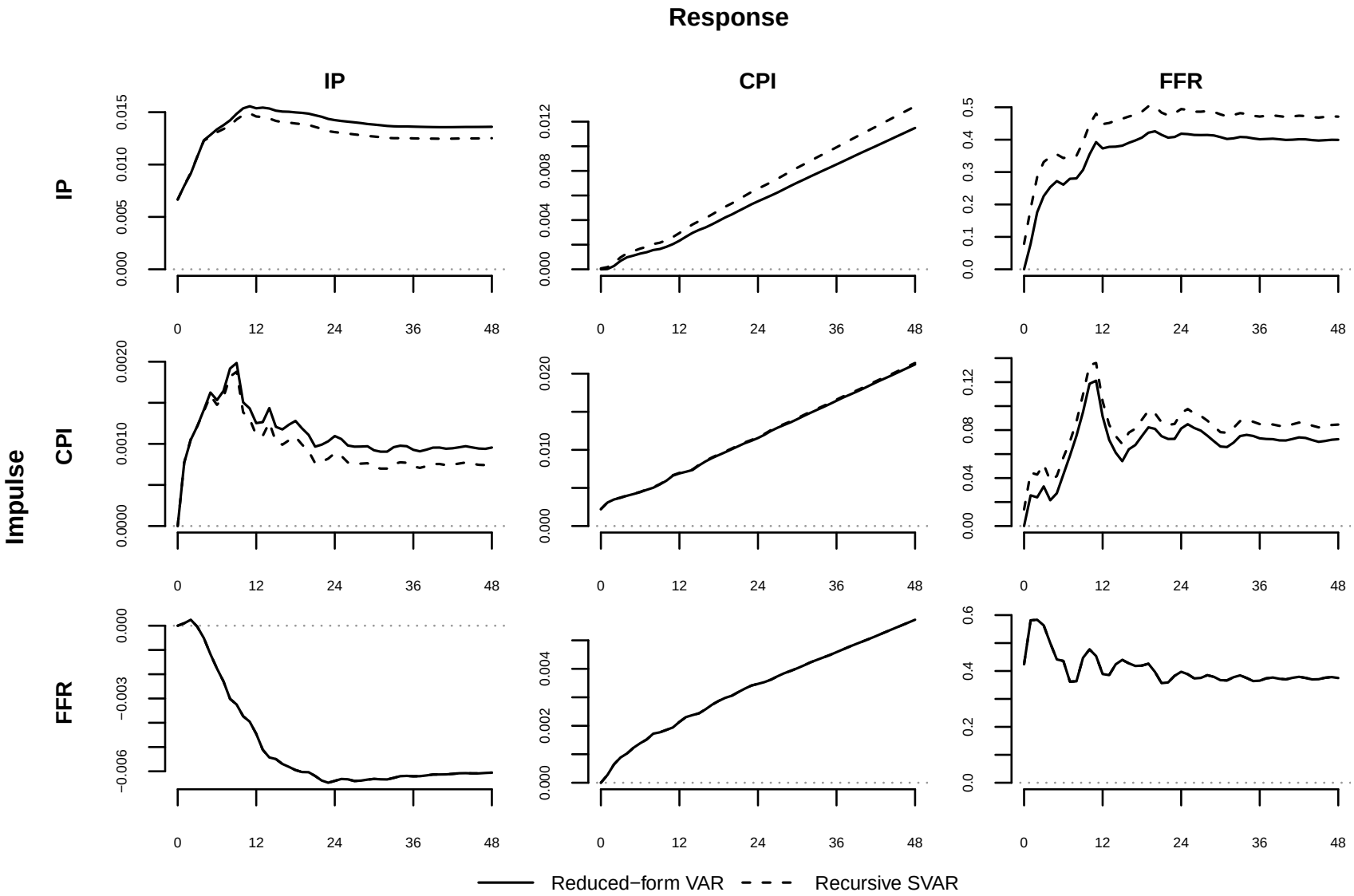
**Shock magnitude:** Each impulse represents a one-standard-deviation structural innovation, with its initial impact determined by the estimated model

**Response units:** Accumulate IP and FFR once, and CPI twice. IP and CPI are plotted in log points; FFR in percentage points

**Ordering comparisons:** Retain one-standard-deviation structural shocks under each recursive ordering

$\Rightarrow$ Initial FFR movements may differ across specifications, since shock magnitudes are determined separately

# Reduced-Form VAR versus Recursive SVAR



# Reduced-Form versus Structural IRFs: Interpretation

**Shock decomposition:** The reduced-form and recursive VAR share the same dynamic coefficients; only the impact vector changes

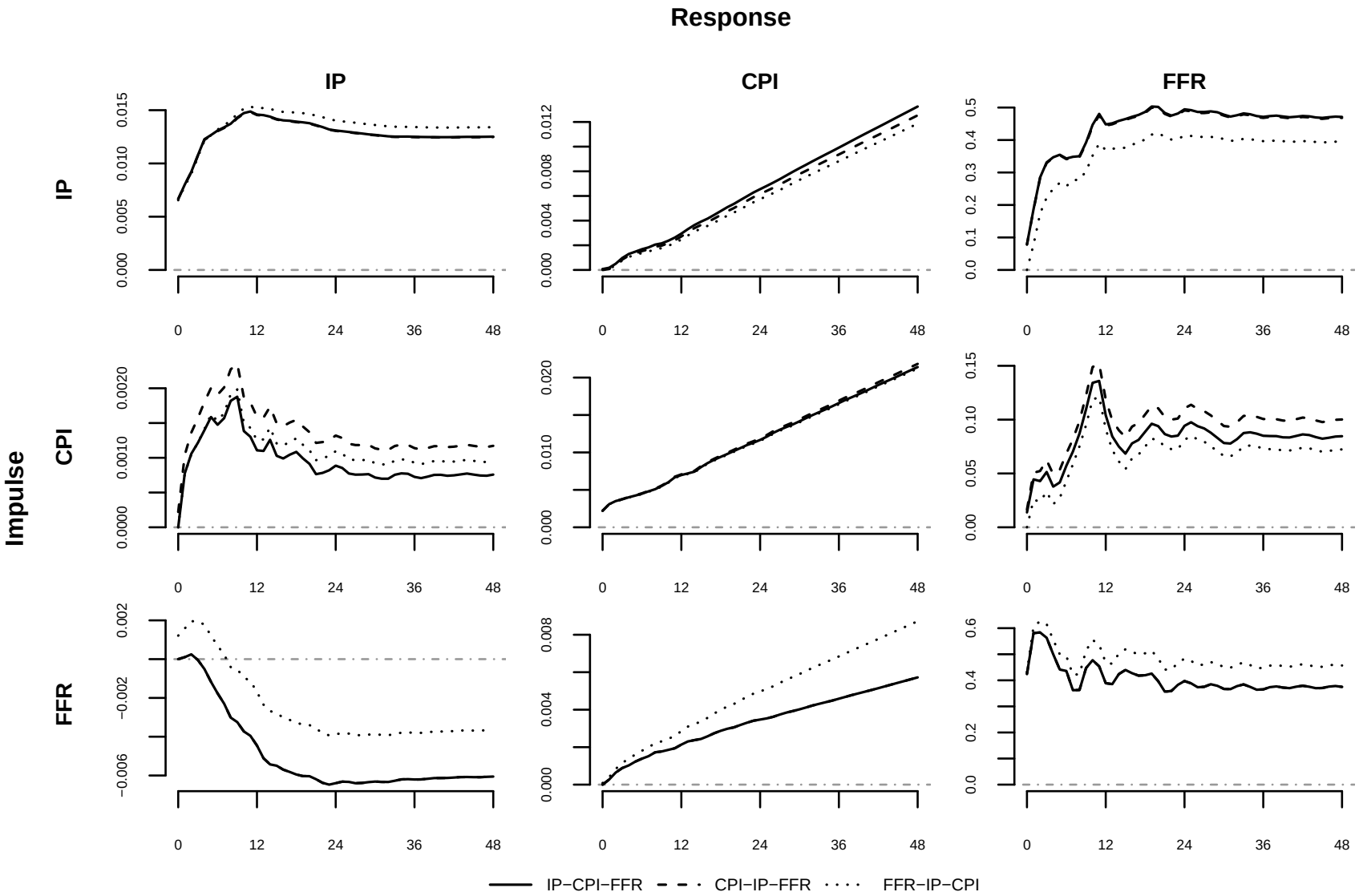
- The IP and CPI rows differ because recursive identification separates correlated contemporaneous innovations
- With FFR ordered last, its structural impact vector is $Pe_3 = p_{33}e_3$

$$\Theta_h e_3 = p_{33}\Phi_h e_3$$

- Thus the FFR response rows coincide once their initial FFR impact is matched
- In the plotted estimates, a positive FFR innovation is followed by falling IP and rising CPI

$\Rightarrow$ Recursive identification gives the policy innovation a structural interpretation, but **does not eliminate the price puzzle**

# Sensitivity to Alternative Recursive Orderings



# Ordering Sensitivity: Interpretation

Changing the variable ordering changes the contemporaneous restrictions imposed on structural shocks

**IP–CPI–FFR versus CPI–IP–FFR:** Both specifications place FFR last and condition its innovation on the same macroeconomic variables

- Their one-SD monetary-policy shock responses coincide, even though the other structural shocks differ

**FFR–IP–CPI:** Placing the policy rate first permits immediate IP and CPI responses while restricting contemporaneous monetary-policy reactions to macroeconomic innovations

- Different identifying restrictions produce different structural IRFs

$\Rightarrow$ The comparison reflects changes in identification assumptions, without imposing a common FFR impact

# VAR/LP IRFs Comparison

# Local Projections: Direct versus Iterated Responses

Local projections (Jordà, 2005) estimate the responses separately at each horizon

$$y_{t+h} = \alpha_h + \Psi_h y_t + \sum_{\ell=1}^{14} \Gamma_{\ell,h} y_{t-\ell} + v_{t+h}^{(h)}.$$

- A separate regression is estimated for each $h = 0, \ldots, 48$
- Current IP, CPI and FFR enter jointly, controlling for the other contemporaneous variables and the lagged history
- In contrast, a VAR obtains all its horizons by repeatedly applying estimated lag matrices

$\Rightarrow$ LPs relax the common-dynamic-system restriction, but may be noisier at long horizons

# VAR versus LP: Estimation Trade-Off

| | **VAR** | **Local projections** |
|---|---|---|
| Estimation | One dynamic model, iterated | Separate regression at each horizon |
| Strength | Smoother, often more precise responses | Flexible response dynamics |
| Weakness | Propagated bias under dynamic misspecification | Greater long-horizon sampling variation |

Both estimators can recover the same population IRFs under appropriate conditions; in finite samples their bias-variance properties differ

$\Rightarrow$ **Do the output response and the price puzzle from Part I depend on direct versus iterated estimation?**

# Reduced-Form LP: Comparable Specification

We retain the same transformed variables, 14 lagged controls and 48-month response horizon as the reduced-form VAR

**Estimation:** Horizon-specific OLS regressions using `lp_lin_iv()` with `use_twosls = FALSE`

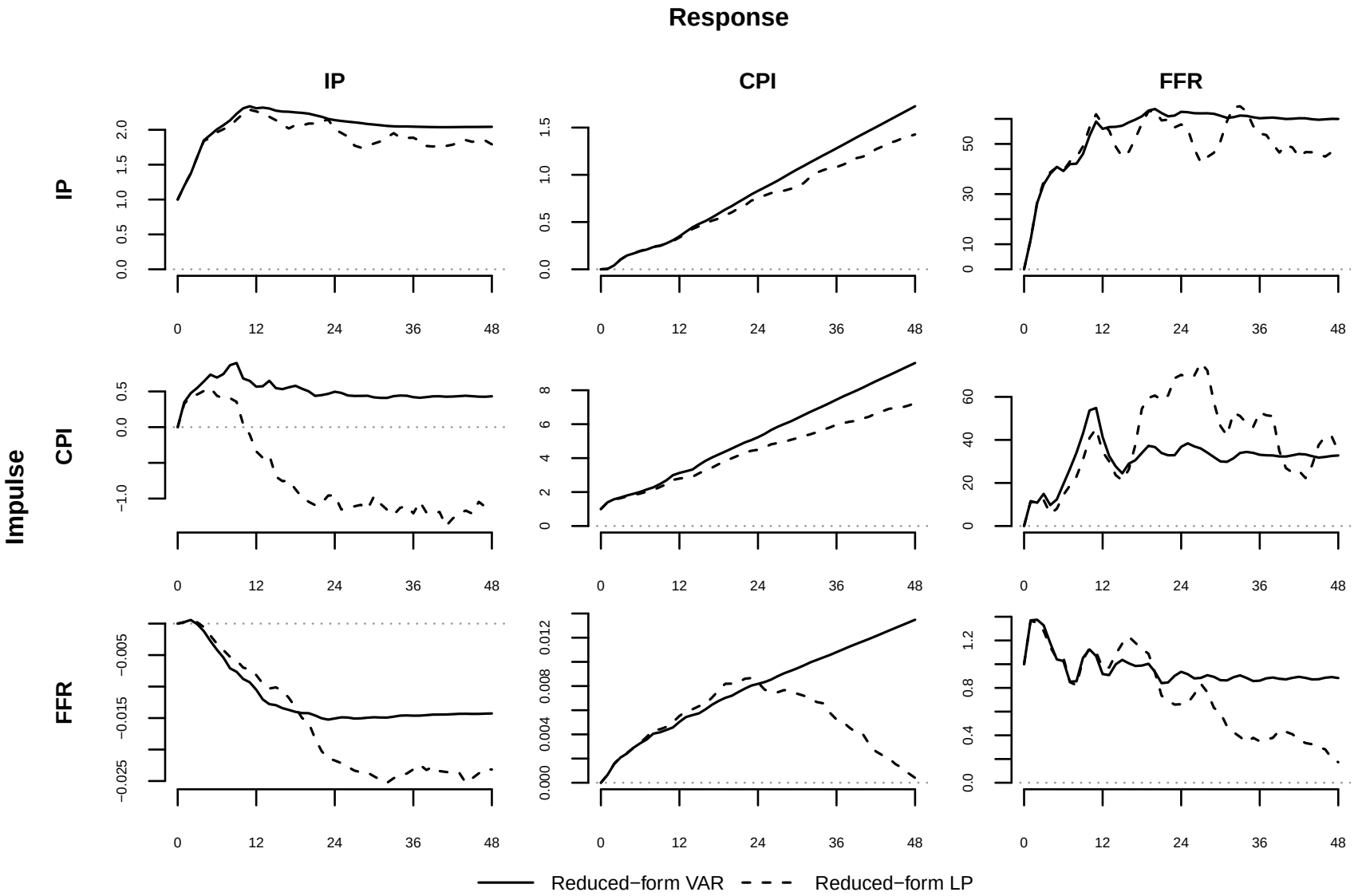
- `shock` selects the contemporaneous impulse variable; `contemp_data` supplies the other two current variables

**Shock definition:** A one-unit conditional innovation in the transformed variable, without Cholesky identification

**Response reconstruction:** Accumulate IP and FFR once, and CPI twice, to obtain level responses

$\Rightarrow$ Comparing these LPs with reduced-form VAR IRFs isolates differences between direct and iterated dynamic estimation

# Reduced-Form IRFs: VAR versus LP



# Reduced-Form VAR versus LP: Interpretation

**On impact:** Both estimators recover the identity response matrix for one-unit reduced-form innovations

**After impact:** VAR imposes a common dynamic model across horizons, whereas each LP coefficient is estimated separately

- FFR-to-IP responses are negative in both methods, with a larger long-run decline under LP
- Positive FFR-to-CPI response indicates a price puzzle initially; the LP response subsequently weakens and turns down relative to the more persistent VAR response
- At longer horizons, visibly different responses are not by themselves evidence that either method is statistically superior

$\Rightarrow$ The persistence of the price puzzle is sensitive to the estimator; structural identification is still unresolved

# Structural IRFs Using Local Projections

# Reduced-Form versus Structural Local Projections

Both approaches use the same horizon-specific OLS estimates, collected in the response matrix $\Psi_h$

**Reduced-form LP:** Measures responses to a one-unit innovation in variable $k$, conditional on the other variables

$$IRF_{k,h}^{RF} = \Psi_h e_k$$

**Structural LP:** Applies the Cholesky impact matrix $P$ from the baseline VAR to identify structural shocks

$$IRF_{k,h}^{S} = \Psi_h P e_k, \qquad \Sigma_u = PP'$$

# Reduced-Form versus Structural Local Projections

**For the monetary-policy shock (FFR ordered last):**

$$Pe_3 = p_{33}e_3 \quad \Rightarrow \quad IRF^S_{FFR,h} = p_{33} IRF^{RF}_{FFR,h}$$

- The FFR response paths have the same shape, but different shock magnitudes
- For IP and CPI shocks, structural identification can combine several reduced-form responses

$\Rightarrow$ **Key distinction:** Same LP estimation, different shock identification and normalization

# Identification Strategies for Structural LPs

Local projections estimate dynamic responses directly, but causal interpretation still requires identifying the structural disturbance

**Recursive restrictions:** Impose contemporaneous timing assumptions through an ordered impact matrix

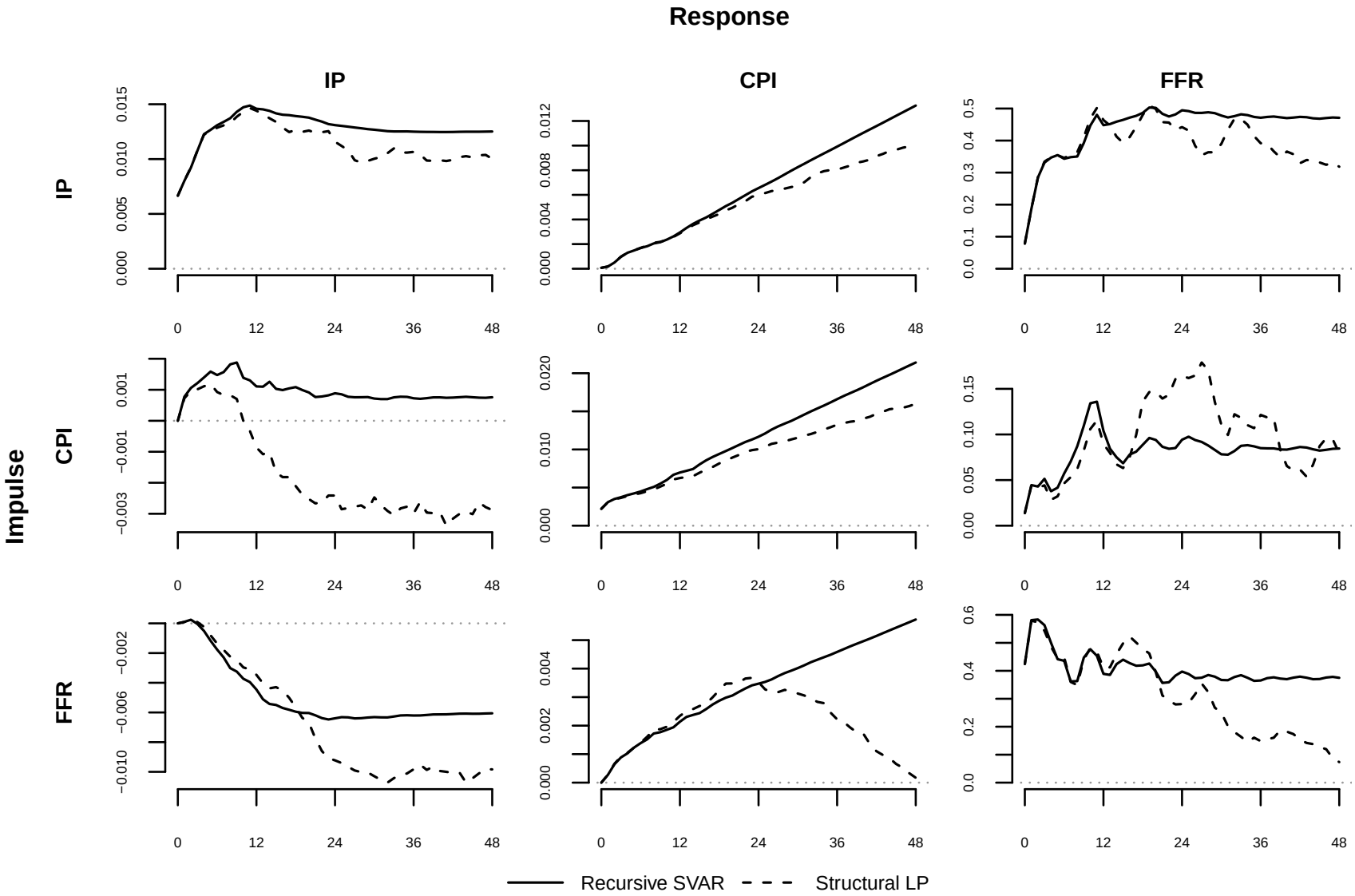
- Example: IP and CPI cannot respond immediately to an unexpected FFR shock

**External instruments (LP-IV):** Use an instrument correlated with the policy shock but exogenous to other structural disturbances

**Sign or long-run restrictions:** Identify shocks by restricting selected responses rather than relying only on contemporaneous zeros

$\Rightarrow$ Here: Use the baseline Cholesky restrictions so that the structural VAR and LP differ in estimation, not identification

# Structural IRFs: Recursive SVAR versus LP



# Structural LP versus SVAR: Interpretation

**Same identification:** Both specifications use IP–CPI–FFR Cholesky restrictions and identical structural impact vectors at horizon zero

**Policy transmission:** The FFR impulse row isolates the comparison of monetary-policy IRFs

- Both estimates imply a negative response of IP after the initial tightening, but LP becomes more negative at longer horizons
- Both initially imply a positive CPI response; LP prices weaken later, while VAR prices continue to rise in the plotted estimates
- FFR persistence also differs as the horizons lengthen

$\Rightarrow$ VAR-LP differences measure estimator sensitivity; neither approach independently validates the recursive shock identification

# Shock Proxies

# Why Introduce MP1?

The recursive SVAR identifies monetary-policy shocks from the FFR innovation after accounting for contemporaneous IP and CPI developments

But the Fed may respond to information about future economic activity that is absent from the VAR

- Consequently, the remaining FFR innovation need not be a pure, exogenous policy disturbance
- High-frequency surprises around FOMC announcements provide another measure of unexpected policy news

The question changes from

**What follows an FFR residual shock? $\Rightarrow$ What follows an unexpected MP1 innovation?**

# What Does MP1 Measure?

MP1 measures the unexpected current-meeting policy-target component inferred from financial-market changes around FOMC announcements

**Timing:** A narrow announcement window limits exposure to unrelated economic news, without guaranteeing exogeneity

**Data:** Monthly MP1 observations are available from February 1994 to December 2019 (311 observations)

**Role:** MP1 is an observable proxy for policy news, not an observed structural monetary-policy disturbance

$\Rightarrow$ We use the unexpected component of MP1 to trace the macroeconomic response to policy-related announcements

# Internal versus External Instruments

The distinction is where the proxy enters the empirical system

**External instrument:** MP1 stays outside the three-variable VAR. Proxy–residual covariance moments or LP-IV are used to identify policy responses

**Internal instrument:** MP1 enters the VAR itself as the first variable, and its first orthogonalized innovation becomes the impulse of interest

- We implement the second approach, no external-instrument 2SLS estimation is performed

$\Rightarrow$ The proxy is not just a control, it supplies the innovation whose macroeconomic effects we study

# How MP1 Enters the Augmented VAR

Add MP1 first, ahead of industrial production, consumer prices and the policy rate:

$$W_t = \begin{pmatrix} MP1_t \\ \Delta \log IP_t \\ \Delta^2 \log CPI_t \\ \Delta FFR_t \end{pmatrix}, \qquad W_t = c + \sum_{\ell=1}^{9} A_\ell^W W_{t-\ell} + v_t.$$

The four-variable VAR is estimated by OLS. Its reduced-form covariance matrix is decomposed as $\Sigma_W = P_W P_W'$:

$$P_W = \begin{pmatrix} p_{11} & 0 & 0 & 0 \\ p_{21} & p_{22} & 0 & 0 \\ p_{31} & p_{32} & p_{33} & 0 \\ p_{41} & p_{42} & p_{43} & p_{44} \end{pmatrix}.$$

$\Rightarrow$ Because MP1 is ordered first, its identified innovation can move IP, CPI and FFR immediately

# Which Innovation Identifies Monetary Policy?

The MP1-SVAR has two different innovations that must not be confused

**First innovation – MP1 (the shock we use):**

$$IRF_h^{MP1} = \Phi_h^W P_W e_1, \qquad P_W e_1 = (p_{11}, p_{21}, p_{31}, p_{41})'$$

**Fourth innovation – FFR (not our proxy shock):**

$$IRF_h^{FFR} = \Phi_h^W P_W e_4, \qquad P_W e_4 = (0, 0, 0, p_{44})'$$

- The fourth innovation is the FFR disturbance remaining after conditioning on MP1, IP and CPI

$\Rightarrow$ To study MP1-based identification, shock MP1 ($e_1$), not FFR ($e_4$)

# Proxy Validity: Conditions and Consequences

Let $z_t = MP1_t$ and $\mathcal{H}_{t-1}$ denote past information. A valid proxy innovation must satisfy:

$$\underbrace{z_t^{\perp}}_{z_t - \mathrm{Proj}(z_t|\mathcal{H}_{t-1})} = \alpha \varepsilon_t^{MP} + \eta_t, \qquad \alpha \neq 0$$

**Relevance:** MP1 must contain information about the current monetary-policy shock ($\alpha \neq 0$)

- If weak, estimated responses become imprecise and sensitive to sampling variation

**Exogeneity:** Proxy noise $\eta_t$ must be orthogonal to other structural disturbances

- Fed-information effects may contaminate MP1 with economic news, biasing the interpretation

**Lead–lag exclusion:** The proxy innovation must isolate the policy shock at time $t$, not shocks from other dates

- Violations can distort the timing and persistence of the estimated IRFs

**If these conditions fail, MP1-based responses need not represent causal monetary-policy transmission**

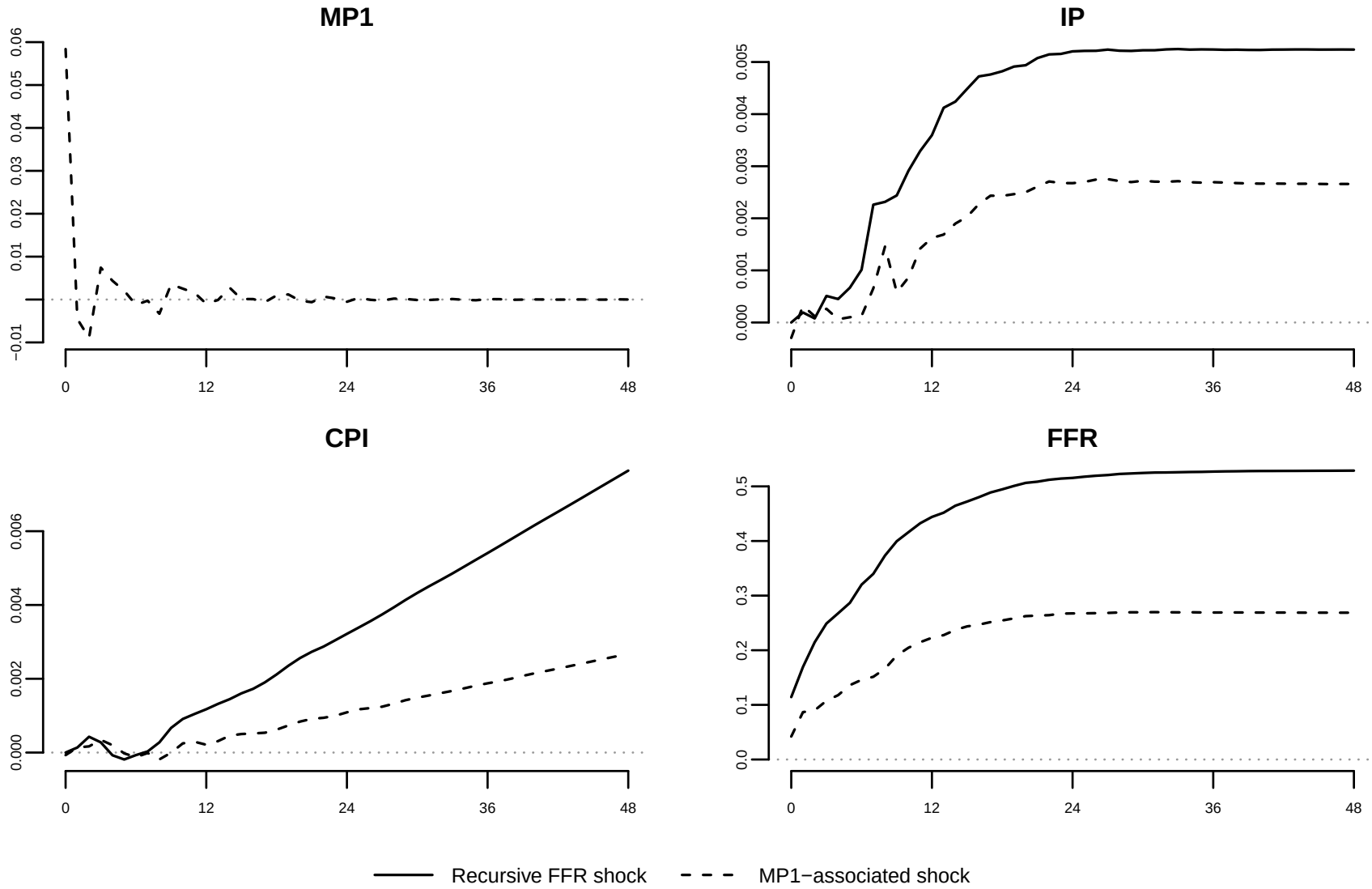
# Evidence on MP1 Relevance and Stability

The conditional MP1 coefficient is statistically informative about its association with unexpected FFR movements. The companion roots indicate a stable estimated VAR

- These diagnostics support empirical relevance but do not establish correlation with the latent policy shock
- They cannot rule out contamination from Fed-information effects or other structural disturbances

| Diagnostic | Estimate |
|---|---|
| Proxy-sample observations | 311 |
| AIC-selected VAR lag order | 9 |
| Effective VAR observations | 302 |
| MP1–FFR residual correlation | 0.391 |
| Conditional MP1 $F$ statistic | 47.51 |
| Maximum VAR root modulus | 0.901 |

# Structural IRFs: FFR Shock versus MP1 Shock



# FFR versus MP1 Innovations: Interpretation

The figure compares macroeconomic responses to two differently identified one-standard-deviation innovations

**MP1:** Only the augmented model reports the proxy's response to its own innovation

**Industrial production:** Both shocks produce positive medium-run IP responses in the common sample, but the recursive response is larger

**Consumer prices:** CPI rises under both specifications; introducing MP1 does not eliminate the price puzzle

**Federal funds rate:** Both shocks raise FFR, but the recursive innovation produces a larger initial policy-rate movement

$\Rightarrow$ Differences in magnitude reflect both the size of the induced policy-rate movement and the estimated transmission dynamics

# Which Identification Is More Trustworthy?

The two methods identify different innovations, even when the sample and VAR lag order are held fixed

**Recursive SVAR:** Offers a transparent benchmark with explicit within-month timing restrictions, but its FFR innovation may contain endogenous policy responses

**MP1 internal instrument:** Uses announcement-time surprises that are more closely targeted at unexpected policy decisions, subject to relevance and exclusion restrictions

**Empirical evidence:** Both specifications display a positive CPI response, while their IP and FFR responses differ substantially

- The current diagnostics do not establish MP1 exogeneity, and the plotted IRFs have no confidence intervals

$\Rightarrow$ Treat MP1 as a valuable identification robustness check, but do not rank its causal estimates above the recursive benchmark without stronger evidence

# Final Recommendations

# Lessons Learned

Our analysis distinguishes two decisions: how to identify economic shocks and how to estimate their dynamic effects

**Identification matters:** Reduced-form innovations are forecast errors, not automatically exogenous economic shocks

**Estimation matters:** Direct LPs and iterated VARs produce different finite-sample response patterns even under the same identification

**The price puzzle remains:** Positive CPI responses appear under recursive and MP1-associated shocks, although their persistence varies across methods

**Proxy identification is conditional:** MP1 introduces information about policy announcements, but relevance and exogeneity remain essential

$\Rightarrow$ Robust policy analysis requires examining both identifying assumptions and dynamic estimation choices

# Final Recommendations

Our preferred approach is to use several complementary specifications rather than rely on one set of impulse responses

**1. Establish a transparent benchmark** Retain the recursive SVAR(14) from Part I, clearly reporting the variable ordering and one-SD shock definition

**2. Check estimator sensitivity** Use structural LPs with the same Cholesky impact matrix to separate estimation differences from identification differences

**3. Examine alternative identification** Compare the recursive SVAR(9) with the MP1-first VAR(9) over a common sample, acknowledging differences in initial FFR impacts

**4. Strengthen inference** Report confidence intervals and investigate instrument validity before drawing causal conclusions from point estimates

$\Rightarrow$ No single specification resolves the price puzzle conclusively; robust conclusions should survive credible alternatives

# Appendix

# Appendix A1: Construction of the MP1 Shock Proxy

**MP1 measures unexpected current-meeting policy-target news** inferred from changes in federal funds futures prices around FOMC announcements

- **Announcement window:** 10 minutes before to 20 minutes after the FOMC statement
- Federal funds futures reflect expectations of the average effective federal funds rate
- The change in the futures-implied rate captures the unexpected policy component:

$$\Delta f_e = f_e^{after} - f_e^{before}$$

# Appendix A1: Construction of the MP1 Shock Proxy

**Adjustment for the timing of the announcement:**

$$MP1_e = \begin{cases} \dfrac{D_e}{D_e - d_e} \Delta f_e^{current}, & D_e - d_e > 7 \\ \Delta f_e^{next}, & D_e - d_e \leq 7 \end{cases}$$

where $D_e$ is the number of days in the month and $d_e$ is the announcement day. For late-month announcements, the next-month futures contract is used

**Monthly aggregation:**

$$MP1_t = \sum_{e \in \mathcal{E}_t} MP1_e$$

- Sum surprises within each month; no-event months are set to zero
- Sample: February 1994–December 2019 (311 monthly observations)

MP1 captures unexpected policy-target news, not the actual policy-rate change or a directly observed structural shock

# Appendix A2: External-Instrument SVAR (Proxy-SVAR)

**Step 1 - Estimate the three-variable reduced-form VAR**

$$y_t = c + \sum_{\ell=1}^{p} A_\ell y_{t-\ell} + u_t, \qquad u_t = P\varepsilon_t$$

**Step 2 - Use MP1 as an external instrument**

MP1 remains outside the VAR. Validity requires:

$$\mathrm{Cov}(MP1_t, \varepsilon_t^{MP}) \neq 0, \qquad \mathrm{Cov}(MP1_t, \varepsilon_t^{j}) = 0 \quad (j \neq MP)$$

**Step 3 - Identify and propagate the policy impact vector**

$$g = \mathrm{Cov}(u_t, MP1_t) \propto Pe_{MP}$$

Normalize to a one-percentage-point FFR increase:

$$b^{MP} = \frac{g}{g_{FFR}} \times 1 \text{ pp}, \qquad IRF_h^{IV} = \Phi_h b^{MP}$$

$\Rightarrow$ MP1 identifies the impact vector; the VAR propagates the shock across horizons

# Appendix A3: LP-IV (Instrumental-Variable Local Projections)

**Step 1 - Instrument the policy-rate movement with MP1**

First-stage regression:

$$\Delta FFR_t = \pi MP1_t + \Lambda' X_{t-1} + v_t$$

where $X_{t-1}$ contains lagged macroeconomic controls

**Step 2 - Estimate the response at each horizon using 2SLS**

$$y_{j,t+h} = \alpha_{j,h} + \theta_{j,h} \widehat{\Delta FFR}_t + \Gamma'_{j,h} X_{t-1} + e_{j,t+h}$$

$$h = 0, 1, \ldots, 48, \qquad j \in \{IP, CPI, FFR\}$$

# Appendix A3: LP-IV (Instrumental-Variable Local Projections)

**Step 3 - Construct the impulse-response functions**

- $\theta_{j,h}$ measures the response to an MP1-instrumented one-percentage-point FFR movement
- Estimate a separate regression for each horizon
- Accumulate IP and FFR once, and CPI twice, to reconstruct responses in levels

$\Rightarrow$ MP1 identifies policy-related variation; local projections estimate each horizon directly, without iterating a VAR

# Appendix A4: MP1 Lag-Predictability Test

Some implications of instrument validity can be examined, but the exclusion restriction is not directly observable

**Lag-predictability test (Stock and Watson, 2018):**

$$H_0 : \beta_1 = \cdots = \beta_9 = 0$$

in a regression of $MP1_t$ on nine lags of the three macroeconomic variables

- Joint F-statistic: 1.49; p-value: 0.059
- This does not reject zero predictability at 5%, but it does at 10%

**Limitation:** The conditional internal-instrument approach allows dependence on observed lagged information

$\Rightarrow$ The diagnostic cannot establish that MP1 is uncorrelated with contemporaneous non-policy shocks